| K L Deemed to be University | Department of computer science and Engineering -1 -- KLBCH<br>Course Handout<br>2026-2027, Odd Sem |
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| **Course Title** | **: OPERATING SYSTEMS AND SYSTEMS PROGRAMMING** |
| **Course Code** | **: 25CS2104E** |
| **L-T-P-S Structure** | **: 2-0-2-4** |
| **Pre-requisite** | : 25EC2101E-DIGITAL DESIGN & COMPUTER ARCHITECTURE |
| **Credits** | **: 4** |
| **Course Coordinator** | **: Kotte Anusha** |
| **Team of Instructors** | : Dyuti Banerjee , Y Devi, Y Devi, Y Devi, MUNIRAJU VADLAMUDI, MUNIRAJU VADLAMUDI, MUNIRAJU VADLAMUDI, Hema Kolla, Hema Kolla, Hema Kolla, khaja sk, DHULIPALLA KRISHNA , Anitha Chekuri, LakshmiPrasanna Yeluri , Swathi D, Swathi D, Karumuri Murthy, Gayatri Mantri, Kotte Anusha, Kotte Anusha, Kotte Anusha, borra sivaiah, borra sivaiah, DHARAMSOTU BHEEKYA, DHARAMSOTU BHEEKYA, DHARAMSOTU BHEEKYA, paramewar maddela, paramewar maddela, pavani Chilumula, pavani Chilumula, Dasari ., Dasari ., Dasari ., Raghupathi Manthena, Raghupathi Manthena, Raghupathi Manthena, Soumya Enukonda, Soumya Enukonda, Soumya Enukonda, MEKALA SANJEEV KUMAR |
| **Teaching Associates** | : |
| **Syllabus :** | CO-1: The OS as a Service Layer - Operating system as a layered service abstraction, User Space and Kernel Space, System Calls and Kernel Services, Role of the shell as a user-space program, Command Execution Journey. Linux Architecture and System Software Stack, Systems Programming Fundamentals. CO-2 : Processes and Process Control , Process Abstraction, Process Lifecycle and State Transitions, Creation, Execution, Synchronization and Termination, User-Level Process Scheduling Concepts, Common Process Management Pitfalls. CO-3 : Inter-Process Communication (IPC), Anonymous Pipes, Named Pipes (FIFOs) Signals and Asynchronous Notifications, Signal Handlers and POSIX Signal Programming, Process Groups, Sessions, and Job Control. CO-4 : Memory Management , Virtual Memory as an Operating System Abstraction, Address Translation and Page Tables, Page Faults and Demand Paging, Linux Process Address Space Layout, Dynamic Memory Allocation in Linux, Copy-on-Write and Process Creation, Memory Errors and Operating System Responses, Memory Debugging and Analysis Tools. CO-5 : File Systems, File Abstractions, and File I/O in Linux , The Unix File Abstraction, Inodes, Directory Entries, and File Naming, Virtual File System (VFS), File Descriptors and Open File Management, Linux File I/O System Calls, Buffered and Unbuffered I/O, Memory-Mapped File I/O, ext4 File System Overview. CO-6 :Concurrency and Synchronization, Threads and Processes, POSIX Threads, Race Conditions and Shared Data, Mutual Exclusion Using Mutexes, Condition Variables and Thread Coordination, Counting Semaphores, Deadlocks and Concurrency Hazards, Advanced Synchronization Concepts |
| **Text Books :** | 1.Operating System Concepts by Abraham Silberschatz, Peter Baer Galvin & Greg Gagne, Wiley, 2. Modern Operating Systems by Andrew S. Tanenbaum & Herbert Bos ,Pearson (2022) 3. Advanced Programming in the UNIX Environment by W. Richard Stevens & Stephen A. Rago, Addison-Wesley . |
| **Reference Books :** | 1. Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau & Andrea C. Arpaci-Dusseau, Wisconsin 2023 (1.10 Ed.) , 2. The Linux Programming Interface A Linux and UNIX System Programming Handbook by Michael Kerrisk ,No Starch Press , 3. Computer Systems: A Programmer's Perspective by Randal E. Bryant & David R. O'Hallaron ,Pearson |
| **Web Links :** | 1. https://nptel.ac.in/courses/106106144 2. https://nptel.ac.in/courses/106105214 3. https://pdos.csail.mit.edu/6.S081/ 4.https://pages.cs.wisc.edu/~remzi/OSTEP/ |
| **MOOCS :** | 1. https://www.coursera.org/learn/os-power-user 2. https://www.coursera.org/learn/hands-on-introduction-to-linux-commands-and- |

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|---|---|
| | shell-scripting 3. https://www.coursera.org/learn/bits-operating-systems |
| **Course Rationale :** | This course enables students to understand how applications interact with hardware through operating system abstractions, system calls, processes, threads, memory, and file systems. This PBL approach is centered on building a Unix-style shell, students gain hands-on experience with core kernel mechanisms and industry-standard systems programming tools forming the foundation for advanced studies in embedded systems, cloud infrastructure, scalable system design, and systems software engineering careers. |
| **Course Objectives :** | Operating Systems and Systems Programming teaches what every senior software engineer is expected to know — what really happens between a program and the hardware. The PBL track is OUTSIDE-IN: students see a working Unix-style shell on day one (their own Shellforge running in a Linux terminal, executing pipelines, redirecting I/O, juggling background jobs) and across twelve weeks they understand every syscall the shell makes — fork, exec, pipe, wait, kill — and the kernel mechanism behind each. The toolchain is the standard Unix toolkit: gcc, make, gdb, valgrind, strace. The course is the prerequisite for embedded software (ESDT, T06), cloud infrastructure (CDIS, T05), system design for scalability (SDSR, T06), and every course that touches systems internals. |

**COURSE OUTCOMES (COs):**

| CO NO | Course Outcome (CO) | PO/PSO | Blooms Taxonomy Level (BTL) |
|---|---|---|---|
| CO1 | Analyze the operating system as a layered service abstraction — kernel mode vs user mode, the syscall interface, the role of the shell as a user-space program — and trace a simple command from typing it to its completion. | PO2,PO1 | 4 |
| CO2 | Apply Unix process-control primitives — fork, exec, wait, exit — to write systems-programming code in C that creates, controls, and reaps child processes correctly. | PO1,PO3,PO5,PSO1 | 3 |
| CO3 | Apply inter-process communication mechanisms — anonymous pipes, named pipes (FIFOs), Unix domain sockets, signals, and shared memory — to design simple multi-process applications. | PO1,PO3,PSO1 | 3 |
| CO4 | Analyze virtual memory — paging, address translation, the TLB, page faults, demand paging, copy-on-write — and explain at the system level what happens when a process accesses memory. | PO1,PO2 | 4 |
| CO5 | Analyze file-system abstractions — inodes, directory entries, the virtual file system layer, mount points, read/write/seek semantics — and use the file-I/O syscalls to read and write structured data robustly. | PO1,PO3,PSO1 | 4 |
| CO6 | Analyze concurrency primitives — POSIX threads, mutexes, condition variables, counting semaphores — to write correct concurrent C programs and reason about race conditions, deadlocks, and starvation. | PO1,PO3,PO4,PSO1 | 4 |

**COURSE OUTCOME INDICATORS (COIs)::**

| Outcome No. | Highest BTL | COI-1 | COI-2 | COI-3 | COI-4 |
|---|---|---|---|---|---|
| CO1 | 4 | **Btl-1** Remember the fundamental components of an operating system, | **Btl-2** Understand the layered service abstraction of an operating system and | **Btl-3** Apply the execution flow of a simple command by tracing its path from the shell | **Btl-4** Analyze the roles of the shell, system call interface, and kernel in processing and |

| Outcome No. | Highest BTL | COI-1 | COI-2 | COI-3 | COI-4 |
|---|---|---|---|---|---|
| | | including kernel mode, user mode, system calls, and the shell. | user-space programs and kernel services. | to the kernel and back to the user. | completing a user command. |
| CO2 | 3 | **Btl-1** Remember the purpose and functionality of Unix process-control primitives. | **Btl-2** Understand the role of Unix process-control primitives in a Unix-based operating system. | **Btl-3** Apply Unix process-control primitives to develop C programs for process control | |
| CO3 | 3 | **Btl-1** Remember the characteristics and usage of inter-process communication mechanisms | **Btl-2** Illustrate the operation and communication flow of pipes, Unix domain sockets, signals. | **Btl-3** Apply inter-process communication mechanisms including pipes, Unix domain sockets, signals, and shared memory multi-process applications. | |
| CO4 | 4 | **Btl-1** Remember the fundamental concepts of virtual memory, including paging, address translation. | **Btl-2** Understand the principles of virtual memory management and describe the roles of paging, address translation. | **Btl-3** Apply the process of virtual-to-physical address translation and apply virtual memory concepts for page fault handling scenarios. | **Btl-4** Analyze virtual memory mechanisms including paging, address translation, TLB operation, page faults, demand paging, and copy-on-write. |
| CO5 | 4 | **Btl-1** Remember the fundamental file-system abstractions in Linux-based operating systems. | **Btl-2** Understand the organization and functionality of file-system abstractions. | **Btl-3** Apply the use of file-I/O system calls to create, access, read, write, and manage structured data in files. | **Btl-4** Analyze file-system abstractions including inodes, directory entries, the VFS layer to design robust file-handling solutions. |
| CO6 | 4 | **Btl-1** Remember the basic concurrency primitives and recognize common concurrency issues. | **Btl-2** Understand the operation of POSIX threads, mutexes, condition variables, and counting semaphores. | **Btl-3** Apply concurrency primitives including POSIX threads, mutexes, condition variables, and counting semaphores. | **Btl-4** Analyze concurrent C programs using POSIX threads, mutexes, condition variables, and counting semaphores to detect race conditions, deadlocks, and starvation scenarios. |

**PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)**

| Po No. | Program Outcome |
|---|---|

**Lecture Course DELIVERY Plan:**

| Sess.No. | CO | COI | Topic | Book No[CH No] [Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| 1 | CO1 | COI-1 | Introduction to OS as a Service Layer | TB1 – CH 1 – Pg 3-25, TB2 – CH 1 – Pg 1-30 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),End Semester Exam,Hackathon,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration |
| 2 | CO1 | COI-4 | User Space and Kernel Space ,Privilege Levels , Mode Switching | TB1 – CH 2 – Pg 61-85, TB2 – CH 1 – Pg 31-48 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Hackathon,In-Term Practical Examination,Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration |
| 3 | CO1 | COI-4 | System Calls and Kernel Services,Types of System Calls, System Call Interface | TB1 – CH 2 – Pg 86-115, TB3 – CH 1 – Pg 1-32 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive) |
| 4 | CO1 | COI-4 | Shell as a User-Space Program , Command Parsing and Execution Flow | TB2 – CH 1 – Pg 48-75, TB3 – CH 8 – Pg 265-285 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive) |
| 5 | CO2 | COI-2 | Process Abstraction and the Concept of Execution | TB3 – CH 1 – Pg 1-32, TB3 – CH 8 – Pg 265-286 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),End Semester Exam,Hackathon,In-Term |

| Sess.No. | CO | COI | Topic | Book No[CH No] [Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| | | | | | | Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration |
| 6 | CO2 | COI-3 | Process Lifecycle & State Transitions (Synchronization & Termination) | TB1 – CH 3 – Pg 119-142, TB2 – CH 2 – Pg 77-102 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive) |
| 7 | CO2 | COI-3 | User-Level Process Scheduling (Preemption & Priority) | TB1 – CH 3 – Pg 143-166, TB2 – CH 2 – Pg 103-122 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive) |
| 8 | CO3 | COI-3 | Inter-Process Communication (IPC) Foundations & Anonymous Pipes | TB1 – CH 5 – Pg 223-248, TB2 – CH 2 – Pg 123-145 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),End Semester Exam,Hackathon,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration |
| 9 | CO3 | COI-3 | Differences between Anonymous Pipes and Named Pipes (persistence and scope) | TB1 – CH 6 – Pg 255-275, TB2 – CH 2 – Pg 145-154 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive) |

| Sess.No. | CO | COI | Topic | Book No[CH No] [Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| 10 | CO3 | COI-3 | Signals and Asynchronous Notifications | TB1 – CH 3 – Pg 167-184, TB2 – CH 2 – Pg 155-175 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive) |
| 11 | CO3 | COI-3 | Job Control Mechanics and UNIX Domain Sockets | TB3 – CH 15 – Pg 543-568 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive) |
| 12 | CO3 | COI-3 | Shared Memory and IPC Selection | TB3 – CH 10 – Pg 353-399 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive) |
| 13 | CO4 | COI-4 | Virtual Memory and Address Translation | TB3 – CH 10 – Pg 399-434 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |
| 14 | CO4 | COI-4 | Page Faults and Demand Paging | TB3 – CH 9 – Pg 321-352 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |
| 15 | CO4 | COI-4 | Linux Process Address Space and Dynamic Memory Allocation | TB1 – CH 10 – Pg 433-460, TB2 – CH 3 – Pg 177-205 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |
| 16 | CO4 | COI-4 | Copy-on-Write (COW) and Memory Errors | TB1 – CH 10 – Pg 461-485, TB2 – CH 3 – Pg 205-230 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |

| Sess.No. | CO | COI | Topic | Book No[CH No] [Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| 17 | CO5 | COI-4 | Unix File Abstraction and Linux File System Architecture | TB1 – CH 10 – Pg 486-515, TB2 – CH 3 – Pg 231-255 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),End Semester Exam,Hackathon,In-Term Practical Examination,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration |
| 18 | CO5 | COI-4 | Inodes, VFS and File Descriptor Management | TB3 – CH 7 – Pg 237-264, TB1 – CH 3 – Pg 142-166 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |
| 19 | CO5 | COI-4 | File I/O System Calls and Buffered and Unbuffered I/O | TB3 – CH 7 – Pg 237-264, TB2 – CH 3 – Pg 255-270 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |
| 20 | CO5 | COI-4 | Memory-Mapped Files and ext4 Overview | TB1 – CH 13 – Pg 563-598, TB2 – CH 4 – Pg 273-305 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |
| 21 | CO6 | COI-4 | Threads and Processes and POSIX threads ,Race Conditions | TB3 – CH 3 – Pg 67-104, TB2 – CH 4 – Pg 305-325 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),End Semester Exam,Hackathon,In-Term Practical Examination,MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration |
| 22 | CO6 | COI-4 | Mutexes and Condition variables | TB3 – CH 3 – Pg 67-135 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning |

| Sess.No. | CO | COI | Topic | Book No[CH No] [Page No] | Teaching-Learning Methods | EvaluationComponents |
|---|---|---|---|---|---|---|
| | | | | | | Engagement,End Semester Exam |
| 23 | CO6 | COI-4 | Counting Semaphores and deadlocks | TB3 – CH 14 – Pg 519-541, TB1 – CH 13 – Pg 598-620 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |
| 24 | CO6 | COI-4 | Different Concurrency and Deadlock handling Techniques | TB2 – CH 2 – Pg 123-154, TB3 – CH 11 – Pg 435-482 | Chalk,PPT,Talk | Active Learning Engagement (In-Class),AI-Enabled Adaptive Learning Engagement,End Semester Exam |

**Lecture Session wise Teaching – Learning Plan**

SESSION NUMBER : 1

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | CampusShell – Smart University Computing Lab Management System is a real-world PBL scenario that demonstrates how students interact with Linux systems through the Unix Shell. It helps learners understand command execution, terminal interaction, process monitoring, and the role of the shell as an interface between users and the operating system. The project connects fundamental Operating System concepts to practical laboratory computing environments. | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Introduction to the Operating System as the software layer between applications and hardware. The discussion focuses on how operating systems hide hardware complexity and provide standardized services such as file access, memory allocation, process management, and device control. | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 2

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Chalk | --- NOT APPLICABLE |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| | | | | --- |
| 20 | A banking application must access disk storage securely without allowing users direct hardware access.Design strategey for this objective | 4 | PPT | One minute paper |
| 20 | User mode versus kernel mode, Privileged instructions, Security and stability considerations ,Hardware-enforced protection ,Mode switching , Trap instructions , Controlled access to kernel services, Benefits of isolation. | 3 | PPT | One minute paper |
| 5 | Conclusion and Summary | 1 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 3

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | System Call - Definition and purpose, Kernel service requests, Categories of system calls, Examples of common Linux system calls. Security implications, Resource sharing concerns, Hardware protection mechanisms, OS-mediated access control. | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Linux Kernel Architecture, Monolithic kernel design, Loadable kernel modules, Major kernel subsystems, Device driver framework, Linux Software Stack, Hardware layer, Kernel layer, System libraries, Shells and user applications. | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 4

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | A DevOps engineer investigates why a command-line utility is running slowly. Using strace, the engineer identifies excessive file accesses and optimizes the application. Analyse how to command tracing is done | 4 | PPT | Immediate feedback |
| 20 | Command parsing , Process creation, Program execution ,2. Kernel View Through strace , Observing system calls. File operations, Memory allocation requests, Process management calls. | 3 | PPT | Immediate feedback |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Conclusion and Summary | 2 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 5

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Definition of a process vs. a program, Structure of the Process Control Block (PCB) , Memory layout of a process (Text, Data, Heap, and Stack segments) ,Process isolation and memory virtualization foundations | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Mechanics of process creation using system calls (e.g., fork() and exec()) , The transition from the New state to the Ready queue ,CPU dispatching mechanics and transition to the Running state ,Parent-child process hierarchies and resource inheritance | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 2 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 6

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | The Blocked or Waiting state triggers (I/O requests, event waits) ,Process synchronization foundations (signals, wait queues, and communication) , Normal vs. abnormal termination system calls (e.g., exit(), abort()) ,Resource reclamation and the concepts of Zombie and Orphan processes | 3 | PPT | Case Study |
| 20 | Role of the Short-Term, Medium-Term, and Long-Term schedulers Non-preemptive scheduling algorithms (First-Come First-Served, Shortest Job First) Scheduling performance criteria (CPU utilization, throughput, turnaround time, waiting time) Single-queue vs. multi-queue foundation models | 3 | PPT | Case Study |
| 5 | Conclusion and Summary | 2 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 7

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Preemptive scheduling mechanics and context switching overhead , Round-Robin (RR) scheduling and the impact of time quantum sizes , Priority-based scheduling and Shortest Remaining Time First (SRTF) , Multilevel Queue and Multilevel Feedback Queue scheduling | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Race conditions and the critical section problem, Deadlock conditions (Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait) , Resource starvation and the aging technique solution ., Priority inversion and basic concurrency bugs | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 8

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Using a terminal command like cat names.txt \| grep "Alice". The output of the cat process is piped directly as the input to the grep process through an anonymous pipe created by the shell. | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Introduction to unidirectional data channels via Anonymous Pipes , System calls for pipe creation and management (e.g., pipe()) , Data flow mechanics: duplicating file descriptors using dup2(),Lifecycle of anonymous pipes within parent-child process hierarchies | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 9

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | A local logging service daemon running on a Linux server. Multiple independent background applications write their error logs to a shared named pipe located at /tmp/app_log.fifo, which the logging daemon continuously reads and processes. | 3 | PPT | Case Study Based Learning |
| 20 | Creating named pipes in the filesystem using mkfifo() or the command line Bi-directional communication patterns using | 3 | PPT | Case Study Based Learning |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| | multiple FIFOs Non-blocking I/O (O_NONBLOCK) behavior with FIFO open operations Access control, permissions, and security considerations for filesystem-based pipes | | | |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 10

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Core synchronous vs. asynchronous kernel notifications , Standard POSIX signals (e.g., SIGKILL, SIGTERM, SIGINT, SIGSEGV), Default OS actions for signals (Terminate, Core Dump, Stop, Ignore).,The lifecycle of a signal: Generation, Pending status, and Delivery | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Session creation using setsid() for process isolation , Controlling terminals (/dev/tty) and their relationship to sessions,, Background vs. Foreground process groups within a session , Creation and structure of Daemon processes (detaching from controlling terminals) | 3 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 11

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Job control signals: SIGTSTP (terminal stop), SIGSTOP (mandatory stop), SIGCONT (continue), Handling orphan process groups and background I/O attempts (SIGTTIN, SIGTTOU), The role of the shell as a session leader mediating terminal access | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | UNIX Domain Sockets in System Services, Usage in Daemons and Background Services, Systemd Socket Activation, Docker and Container Runtime Communication, Real-World Applications and Case Studies, Best Practices for UNIX Domain Socket Programming, Common Design Patterns, Performance Optimization Techniques, Limitations and Challenges of UNIX Domain Sockets, Testing and Validation of Socket-Based Applications. | 3 | PPT | --- NOT APPLICABLE --- |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Conclusion and Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 12

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Shared Memory Concept, Shared Memory Architecture, Creating Shared Memory Segments, Attaching and Detaching Shared Memory, Reading and Writing to Shared Memory | 3 | PPT | Quiz/Test Questions |
| 20 | IPC Performance Considerations, IPC Reliability and Error Handling, IPC Design Challenges, Comparison of Shared Memory and Message Passing, Common IPC Use Cases and Applications. | 3 | PPT | Quiz/Test Questions |
| 5 | Conclusion and Summary | 2 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 13

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | A video-editing application needs to process a 20 GB video file on a computer with only 8 GB RAM. Analyse how can the operating system allow the application to run smoothly without requiring all data to be loaded into physical memory at once | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Virtual Memory , Address space abstraction, Benefits of virtualization , Memory protection , Logical versus physical addresses ,Page tables, MMU operation. | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 14

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| | | | | --- |
| 20 | An online gaming application requires fast access to memory while ensuring security and isolation from other running programs. Analyze how does the hardware automatically translate virtual addresses into physical addresses during every memory access without slowing down execution significantly | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Page Fault Handling, Minor and major faults, Fault detection, OS response, Demand Paging, Lazy loading, Working set concept, Performance implications. | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 15

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | A Linux-based food delivery server handles thousands of customer orders simultaneously. Each order runs as a process with its own address space for code, data, stack, and heap, while dynamic memory allocation is used to create and manage customer, restaurant, and order details at runtime. Efficient memory management ensures the server can scale without crashes or memory leaks. | 4 | PPT | One minute paper |
| 20 | Process Memory Layout ,Text segment,Data segment,Heap,Stack,Dynamic Memory Allocation,malloc() , calloc(), realloc() , free(). | 4 | PPT | One minute paper |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 16

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | A cloud server creates thousands of child processes using fork(). Discuss how Copy-on-Write reduces memory consumption and improves performance by delaying page copying until modification occurs. Explain page sharing, page faults, and COW mechanism. | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Memory Errors in Linux: Buffer Overflow, Dangling Pointers, Memory Leaks, Use-after-Free, Double Free Errors. | 4 | PPT | --- NOT APPLICABLE --- |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 17

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | A storage company stores millions of files. Users can rename files and move them across directories without changing the actual file content ,Analyze how Linux keep track of file metadata and locate files efficiently | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Introduction to Unix File Abstraction, files as objects, file descriptors, metadata. | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 18

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Introduction to Inodes, inode structure, metadata, inode numbers, and file attributes. | 3 | PPT | --- NOT APPLICABLE --- |
| 20 | Directories and File Naming: Directory entries, mapping filenames to inode numbers, hierarchical directory structure, path resolution, hard links and symbolic links, commands (ls -i, stat, ln). | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 19

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 20 | VFS Architecture: Need for VFS, VFS layer components, Superblock, Inode, Dentry, File Objects, interaction between kernel and filesystem drivers,Common system calls (open(), read(), write(), close(), stat()) | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Mounting filesystems, mount hierarchy, root filesystem (/), mounting USB drives and network filesystems, /etc/fstab overview | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 20

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Memory-Mapped Files, Concept of File Mapping, Virtual Memory and Address Space Mapping, Mapping Files into Process Memory, Shared Memory Mapping, Private Memory Mapping, Demand Paging in Memory-Mapped Files, Page Fault Handling | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Evolution from ext2 and ext3, ext4 Architecture, File System Layout, Superblock Structure, Block Groups and Allocation, Inodes and Metadata Management, Extents and Extent Trees, Directory Organization and Indexing, Journaling Mechanism, Delayed Allocation, Multiblock Allocation | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 21

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Threads vs Processes, Process Control Block (PCB), Thread Control Block (TCB), Process Creation and Termination, Thread Creation and Termination, Process States and Transitions, Thread States and Lifecycle, Multithreading Concepts, | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Race Conditions and Critical Sections, Causes of Race Conditions, Examples of Race Conditions, Data Inconsistency Problems, Shared Variable Access Issues, Mutual Exclusion Principles, Critical Section Requirements, Atomic Operations, Synchronization Mechanisms, Mutex Locks, Semaphores, Monitors, Condition Variables, | 4 | PPT | --- NOT APPLICABLE --- |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| | Deadlock in Threaded Programs, Starvation and Livelock, Detecting Race Conditions, | | | |
| 5 | Conclusion and Summary | 2 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 22

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Mutex Fundamentals, Mutex Lock and Unlock Operations, Types of Mutexes, Binary Mutex and Recursive Mutex, Thread Safety Using Mutexes, Deadlocks Associated with Mutexes, Deadlock Prevention Techniques, Performance Considerations of Mutexes | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Condition Variables Overview, Signaling and Waiting Mechanisms, Wait Operations on Condition Variables, Signal and Broadcast Operations, Producer–Consumer Problem Using Condition Variables, Coordination Between Threads, Avoiding Busy Waiting | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 2 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 23

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Semaphore Concept and Operations, Binary Semaphores vs Counting Semaphores, Structure of a Counting Semaphore, Wait (P) Operation, Signal (V) Operation, Resource Counting using Semaphores, Semaphore Implementation Techniques, Critical Section Management with Counting Semaphores, | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | Deadlock Characterization, Deadlock Prevention Techniques, Deadlock Avoidance Concepts, Safe and Unsafe States, Banker's Algorithm, Deadlock Detection Algorithms, Recovery from Deadlocks, Process Termination for Recovery, Resource Preemption Strategies, Deadlock Handling Approaches in Operating Systems, Starvation and Aging | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 24

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 5 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 20 | Producer–Consumer Problem using Counting Semaphores, Reader–Writer Synchronization using Semaphores, Advantages and Limitations of Counting Semaphores, Common Synchronization Errors, Race Conditions and Semaphore Misuse | 4 | PPT | --- NOT APPLICABLE --- |
| 20 | System Model for Deadlocks, Necessary Conditions for Deadlock (Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait), Resource Allocation Graph (RAG) | 4 | PPT | --- NOT APPLICABLE --- |
| 5 | Conclusion and Summary | 4 | Talk | --- NOT APPLICABLE --- |

**Tutorial Course DELIVERY Plan:**
NO Delivery Plan Exists

**Tutorial Session wise Teaching – Learning Plan**

No Session Plans Exists

**Practical Course DELIVERY Plan:**

| Tutorial Session no | Topics | CO-Mapping |
|---|---|---|
| 1 | To apply OS as a service layer abstraction and install Linux Architecture and System Software Stack | CO1 |
| 2 | To apply User Space, Kernel Space concepts and implement System Calls | CO1 |
| 3 | To analyze Processes and Process Control during process life cycle using strace | CO2 |
| 4 | To apply Process Synchronization techniques and implement creation and Termination | CO2 |
| 5 | To implement Anonymous Pipes for IPC using pipe() syscall, the read-end and write-end file descriptors | CO3 |
| 6 | To execute Named pipes (FIFOs): mkfifo() and common signals (SIGINT, SIGTERM, SIGCHLD, SIGSEGV, SIGPIPE) | CO3 |
| 7 | To analyze Virtual Memory and Process Address Space via page tables; the TLB as a cache of recent translations | CO4 |
| 8 | To create dynamic Memory Allocation and Copy-on-Write using malloc() and | CO4 |

| Tutorial Session no | Topics | CO-Mapping |
|---|---|---|
| | free() as user-space wrappers around brk()/sbrk() and mmap() | |
| 9 | To analyze File Descriptors and Linux File I/O, regular files, directories, devices, sockets, FIFOs, symbolic links. | CO5 |
| 10 | To implement Inodes, Directories, and Memory-Mapped Files using open(), read(), write(), lseek(), close(): the canonical file-I/O syscalls | CO5 |
| 11 | To execute POSIX Threads using pthread_create, pthread_join, pthread_exit; thread-local storage and Race Conditions | CO6 |
| 12 | To implement Semaphores, Condition Variables, and Deadlock Analysis using pthread_mutex_t; lock/unlock, pthread_cond_t; wait/signal/broadcast | CO6 |

**Practical Session wise Teaching – Learning Plan**

SESSION NUMBER : 1

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and Viva voce | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 2

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to | 3 | PPT | --- NOT APPLICABLE --- |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| | another. Explain how control transitions between user space and kernel space during execution. | | | |
| 30 | Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 3

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations. | 3 | Talk | --- NOT APPLICABLE --- |
| 10 | Result Analysyis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 4

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques. | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 5

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency. | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Develop a program that executes the equivalent of the shell command: ls -l \| grep ".c" using fork(), pipe(), dup2(), and exec() system calls. | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 6

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Create a client-server application using Named Pipes (FIFOs). The client sends messages and the server processes and responds to them. Analyze FIFO behavior when multiple clients communicate simultaneously. | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers. | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 7

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 8

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Develop a program using malloc(), calloc(), realloc(), and free(). Monitor memory allocation behavior and identify memory leaks using Valgrind. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Develop a program that demonstrates Copy-on-Write behavior after a fork(). Analyze memory usage before and after modifying data in the child process. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 9

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite). | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Develop a program that redirects standard input and output using file descriptors and dup2(). Demonstrate how shells implement I/O redirection. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 10

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Investigate inode structures using commands such as ls -i, stat, and find. Create hard links and symbolic links and explain their impact on inode allocation. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Design a program using mmap() to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE |

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| | | | | --- |

SESSION NUMBER : 11

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Develop a multithreaded counter application using POSIX threads (pthread_create, pthread_join). Demonstrate race conditions by allowing multiple threads to update a shared variable concurrently. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Modify the previous program using mutex locks (pthread_mutex_lock, pthread_mutex_unlock) and compare the outputs with the race-condition version. Analyze the impact of synchronization. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 12

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Implement the Producer-Consumer problem using counting semaphores and POSIX threads. Evaluate synchronization correctness and system throughput under different buffer sizes. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Design a deadlock scenario involving multiple threads competing for shared resources. Demonstrate deadlock occurrence, identify the four necessary conditions for deadlock, and implement a deadlock prevention strategy using resource ordering. | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE --- |

**Skilling Course DELIVERY Plan:**

| Skilling session no | Topics/Experiments | CO-Mapping |
|---|---|---|
| 1 | To analyze Project Orientation, Linux Development Environment Setup and Shell Architecture Overview | CO1 |

| Skilling session no | Topics/Experiments | CO-Mapping |
|---|---|---|
| 2 | To designing the Overall Shell Control Flow and Main Event Loop and give the structure for the project | CO1 |
| 3 | To implement Arrow Key Navigation for Command History Recall , Dynamic Memory Management Strategies for Shell Data Structures | CO1 |
| 4 | To analyze tokens Command Parsing and Lexical Analysis, Building a Robust Parser Supporting Unix Command Syntax Rules | CO1 |
| 5 | To Implement Single Quote Handling for Literal String Preservation | CO2 |
| 6 | To Implement Backslash Escaping Mechanisms for Special Character Processing | CO2 |
| 7 | To implement Process Synchronization and Child Process Monitoring Using waitpid() | CO2 |
| 8 | To Implement Environment Variable Expansion During Command Parsing | CO2 |
| 9 | To Implement the cd Built-in Command with Directory State Management | CO3 |
| 10 | To Implement pwd and exit Built-in Commands for Shell State Management | CO3 |
| 11 | To design and Implement of Command History Storage and Retrieval System | CO3 |
| 12 | To implement Two-Stage Command Pipelines Using pipe() and dup2() | CO3 |
| 13 | To implement input Redirection Using Standard Input Stream Replacement | CO4 |
| 14 | To append Mode Output Redirection and File Growth Management | CO4 |
| 15 | To implement Error Stream Redirection and Standard Error Management Techniques | CO4 |
| 16 | To implement Background Process Execution and Job Creation Using Command Ampersands | CO4 |
| 17 | To Implement the jobs Built-in Command for Active Job Inspection ,Foreground Job Control Implementation Using fg Command Functionality | CO5 |
| 18 | To implement Background Job Resumption and Process Continuation with bg Command,Unix Signal Fundamentals and Signal Delivery Mechanisms in Interactive Shells | CO5 |
| 19 | To implement Signal Handler Development Using sigaction() for Reliable Signal Processing | CO5 |
| 20 | To implement Handling Ctrl-Z Stop Signals and Process Suspension Mechanisms, | CO5 |
| 21 | To implement Integration of Parsing, Execution, Redirection, and Pipeline Subsystems | CO6 |

| Skilling session no | Topics/Experiments | CO-Mapping |
|---|---|---|
| 22 | To implement Memory Leak Detection and Resource Cleanup Using Valgrind Analysis | CO6 |
| 23 | To implement Stress Testing ShellForge Under High Process and Pipeline Workloads | CO6 |
| 24 | End-to-End Demonstration Preparation Covering Pipelines, Redirection, Job Control, and Signals | CO6 |

**Skilling Session wise Teaching – Learning Plan**

SESSION NUMBER : 1

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 2

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 3

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 4

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 5

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 6

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 7

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | Process Synchronization and Child Process Monitoring Using waitpid() | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 8

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 9

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 3 | Talk | --- NOT APPLICABLE --- |
| 30 | To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 10

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 11

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Store Commands, Maintain History Buffer, Handle Capacity Limits, Retrieve Entries, Display History, Validate Consistency | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Create Pipeline Structures, Store Commands, Maintain Execution Order, Connect Processes, Validate Pipeline Layout, Test Configurations | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 12

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Create Pipes, Redirect Streams, Execute Connected Commands, Close Descriptors, Synchronize Processes, Test Data Flow | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Support Multiple Pipes, Manage Descriptor Arrays, Launch Multiple Processes, Coordinate Execution, Cleanup Resources, Test Long Pipelines | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 13

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 14

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 15

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Parse Combined Redirection, Duplicate File Descriptors, Merge Streams, Verify Ordering, Handle Edge Cases, Test Integrated Outputs | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Parse Complex Commands, Combine Pipes and Redirections, Resolve Execution Order, Validate Syntax, Generate Execution Plans, Test Scenarios | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | Talk | --- NOT APPLICABLE --- |

SESSION NUMBER : 16

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Detect Background Jobs, Launch Non-Blocking Processes, Return Prompt Immediately, Store Job Information, Monitor Execution, Test Background Tasks | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To Create Job Table, Store Process Groups, Track States, Update Status, Remove Completed Jobs, Validate Job Records | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Conclusion and Summary | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 17

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching | 3 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 3 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 18

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 19

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Register Handlers, Configure Signal Actions, Handle Interrupts, Manage Signal Context, Verify Stability, Test Signal Events | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | To Capture SIGINT, Forward Signals, Protect Shell Process, Terminate Foreground Jobs, Update Job States, Test Interrupt Handling | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 20

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Capture SIGTSTP, Suspend Processes, Update Job Table, Preserve State, Resume Later, Test Suspension Workflow | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | To Create Process Groups, Assign Group IDs, Transfer Terminal Control, Restore Ownership, Coordinate Signals, Validate Behavior | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 21

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor Components | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | To Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 22

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 23

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Result analysis and viva | 4 | PPT | --- NOT APPLICABLE --- |

SESSION NUMBER : 24

No Session Outcomes are mapped

| Time(min) | Topic | BTL | Teaching-Learning Methods | Active Learning Methods |
|---|---|---|---|---|
| 10 | Attendance and recap | 1 | Talk | --- NOT APPLICABLE --- |
| 30 | To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review | 4 | PPT | --- NOT APPLICABLE --- |
| 30 | Conclusion and Summary | 3 | PPT | --- NOT APPLICABLE --- |

**WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDEDED PROBLEM-SOLVING EXERCISES etc:**

| Week | Assignment Type | Assignment No | Topic | Details | co |
|---|---|---|---|---|---|

**COURSE TIME TABLE:**

| | Hour | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| Day | Component | | | | | | | | | |
| **Mon** | Theory | --- | --- | -- | -- | H-S12,H-S13 | --- | -- | --- | --- |
| | Tutorial | --- | --- | -- | -- | -- | --- | -- | --- | --- |
| | Lab | --- | --- | H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S12,H-S12,H-S13,H-S13 | H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S12,H-S12,H-S13,H-S13 | -- | --- | -- | --- | --- |
| | Skilling | --- | --- | -- | -- | H-S7,H-S7,H-S8,H-S8,H-S9,H-S9 | --- | H-S7,H-S7,H-S8,H-S8,H-S9,H-S9 | --- | --- |
| **Tue** | Theory | -- | -- | H-S7,H-S8,H-S9 | H-S7,H-S8,H-S9 | -- | --- | --- | --- | --- |
| | Tutorial | -- | -- | -- | -- | -- | --- | --- | --- | --- |
| | Lab | -- | -- | -- | -- | -- | --- | --- | --- | --- |
| | Skilling | H-S4,H-S4,H-S5,H- | H-S4,H-S4,H-S5,H- | -- | -- | H-S10,H-S10,H- | --- | --- | --- | --- |

| | Hour | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| | | S5,H-S6,H-S6 | S5,H-S6,H-S6 | | | S11,H-S11 | | | | |
| Wed | Theory | H-S10,H-S11 | H-S10,H-S11 | -- | -- | -- | -- | -- | --- | --- |
| | Tutorial | -- | -- | -- | -- | -- | -- | -- | --- | --- |
| | Lab | -- | -- | -- | -- | H-S1,H-S1,H-S2,H-S2,H-S3,H-S3 | -- | H-S1,H-S1,H-S2,H-S2,H-S3,H-S3 | --- | --- |
| | Skilling | -- | -- | H-S1,H-S1,H-S2,H-S2,H-S3,H-S3 | H-S1,H-S1,H-S2,H-S2,H-S3,H-S3 | H-S12,H-S12,H-S13,H-S13 | -- | -- | --- | --- |
| Thu | Theory | H-S1,H-S2,H-S3 | H-S1,H-S2,H-S3 | -- | -- | --- | -- | --- | -- | -- |
| | Tutorial | -- | -- | -- | -- | --- | -- | --- | -- | -- |
| | Lab | -- | -- | -- | -- | --- | -- | --- | -- | -- |
| | Skilling | -- | -- | H-S7,H-S7,H-S8,H-S8,H-S9,H-S9 | H-S7,H-S7,H-S8,H-S8,H-S9,H-S9 | --- | -- | --- | H-S10,H-S10,H-S11,H-S11 | H-S10,H-S10,H-S11,H-S11 |
| Fri | Theory | H-S4,H-S5,H-S6 | H-S4,H-S5,H-S6 | --- | --- | --- | -- | -- | -- | -- |
| | Tutorial | -- | -- | --- | --- | --- | -- | -- | -- | -- |
| | Lab | -- | -- | --- | --- | --- | -- | -- | -- | -- |
| | Skilling | -- | -- | --- | --- | --- | -- | H-S12,H-S12,H-S13,H-S13 | H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S10,H-S10,H-S11,H-S11 | H-S4,H-S4,H-S5,H-S5,H-S6,H-S6 |
| Sat | Theory | -- | -- | -- | -- | --- | -- | -- | -- | H-S12,H-S13 |
| | Tutorial | -- | -- | -- | -- | --- | -- | -- | -- | -- |
| | Lab | H-S10,H-S10,H-S11,H-S11 | H-S10,H-S10,H-S11,H-S11 | -- | -- | --- | -- | -- | H-S7,H-S7,H-S8,H-S8,H-S9,H-S9 | H-S7,H-S7,H-S8,H-S8,H-S9,H-S9 |
| | Skilling | -- | -- | H-S1,H-S1,H-S2,H-S2,H-S3,H-S3 | H-S1,H-S1,H-S2,H-S2,H-S3,H-S3 | --- | -- | H-S12,H-S12,H- | H-S12,H-S12,H-S13,H-S13 | -- |

| | Hour | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | S13,H-S13 | | |
| **Sun** | Theory | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Tutorial | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Lab | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| | Skilling | -- | -- | -- | -- | -- | -- | -- | -- | -- |

**REMEDIAL CLASSES:**

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

**SELF-LEARNING:**

Assignments to promote self-learning, survey of contents from multiple sources.

| S.no | Topics | CO | ALM | References/MOOCS |
|---|---|---|---|---|
| 1 | Evolution of Operating Systems and Service Abstractions, | CO1 | --- NOT APPLICABLE --- | 1.Operating System Concepts by Abraham Silberschatz, Peter Baer Galvin & Greg Gagne, Wiley, 2. Modern Operating Systems by Andrew S. Tanenbaum & Herbert Bos ,Pearson (2022) 3. Advanced Programming in the UNIX Environment by W. Richard Stevens & Stephen A. Rago, Addison-Wesley . |
| 2 | Layered Operating System Architecture | CO1 | --- NOT APPLICABLE --- | 1. Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau & Andrea C. Arpaci-Dusseau, Wisconsin 2023 (1.10 Ed.) , 2. The Linux Programming Interface A Linux and UNIX System Programming Handbook by Michael Kerrisk ,No Starch Press , 3. Computer Systems: A Programmer's Perspective by Randal E. Bryant & David R. O'Hallaron ,Pearson |
| 3 | Linux Architecture and System Software Stack | CO2 | --- NOT APPLICABLE --- | 1.Operating System Concepts by Abraham Silberschatz, Peter Baer Galvin & Greg Gagne, Wiley, 2. Modern Operating Systems by Andrew S. Tanenbaum & Herbert Bos ,Pearson |

| S.no | Topics | CO | ALM | References/MOOCS |
|---|---|---|---|---|
| | | | | (2022) 3. Advanced Programming in the UNIX Environment by W. Richard Stevens & Stephen A. Rago, Addison-Wesley . |
| 4 | Different Process scheduling algorithms | CO2 | --- NOT APPLICABLE --- | 1.Operating System Concepts by Abraham Silberschatz, Peter Baer Galvin & Greg Gagne, Wiley, 2. Modern Operating Systems by Andrew S. Tanenbaum & Herbert Bos ,Pearson (2022) 3. Advanced Programming in the UNIX Environment by W. Richard Stevens & Stephen A. Rago, Addison-Wesley . |
| 5 | Common Process Management Pitfalls | CO2 | --- NOT APPLICABLE --- | 1.Operating System Concepts by Abraham Silberschatz, Peter Baer Galvin & Greg Gagne, Wiley, 2. Modern Operating Systems by Andrew S. Tanenbaum & Herbert Bos ,Pearson (2022) 3. Advanced Programming in the UNIX Environment by W. Richard Stevens & Stephen A. Rago, Addison-Wesley . |
| 6 | Signals and Asynchronous Notifications | CO3 | --- NOT APPLICABLE --- | 1.Operating System Concepts by Abraham Silberschatz, Peter Baer Galvin & Greg Gagne, Wiley, 2. Modern Operating Systems by Andrew S. Tanenbaum & Herbert Bos ,Pearson (2022) 3. Advanced Programming in the UNIX Environment by W. Richard Stevens & Stephen A. Rago, Addison-Wesley . |
| 7 | Process Groups, Sessions and Job Control | CO3 | --- NOT APPLICABLE --- | 1. Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau & Andrea C. Arpaci-Dusseau, Wisconsin 2023 (1.10 Ed.) , 2. The Linux Programming Interface A Linux and UNIX System Programming Handbook |

| S.no | Topics | CO | ALM | References/MOOCS |
|---|---|---|---|---|
| | | | | by Michael Kerrisk ,No Starch Press , 3. Computer Systems: A Programmer's Perspective by Randal E. Bryant & David R. O'Hallaron ,Pearson |
| 8 | Linux Process Address Space Layout | CO4 | --- NOT APPLICABLE --- | 1. Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau & Andrea C. Arpaci-Dusseau, Wisconsin 2023 (1.10 Ed.) , 2. The Linux Programming Interface A Linux and UNIX System Programming Handbook by Michael Kerrisk ,No Starch Press , 3. Computer Systems: A Programmer's Perspective by Randal E. Bryant & David R. O'Hallaron ,Pearson |
| 9 | Dynamic Memory Allocation and Memory Debugging | CO4 | --- NOT APPLICABLE --- | 1.Operating System Concepts by Abraham Silberschatz, Peter Baer Galvin & Greg Gagne, Wiley, 2. Modern Operating Systems by Andrew S. Tanenbaum & Herbert Bos ,Pearson (2022) 3. Advanced Programming in the UNIX Environment by W. Richard Stevens & Stephen A. Rago, Addison-Wesley . |
| 10 | Linux File System Internals and VFS | CO5 | --- NOT APPLICABLE --- | 1. Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau & Andrea C. Arpaci-Dusseau, Wisconsin 2023 (1.10 Ed.) , 2. The Linux Programming Interface A Linux and UNIX System Programming Handbook by Michael Kerrisk ,No Starch Press , 3. Computer Systems: A Programmer's Perspective by Randal E. Bryant & David R. O'Hallaron ,Pearson |
| 11 | Advanced Synchronization and Concurrency Hazards | CO6 | --- NOT APPLICABLE --- | 1.Operating System Concepts by Abraham Silberschatz, Peter Baer |

| S.no | Topics | CO | ALM | References/MOOCS |
|---|---|---|---|---|
| | | | | Galvin & Greg Gagne, Wiley, 2. Modern Operating Systems by Andrew S. Tanenbaum & Herbert Bos ,Pearson (2022) 3. Advanced Programming in the UNIX Environment by W. Richard Stevens & Stephen A. Rago, Addison-Wesley . |
| 12 | Deadlock Conditions, Detection, Prevention, and Avoidance | CO6 | --- NOT APPLICABLE --- | 1. Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau & Andrea C. Arpaci-Dusseau, Wisconsin 2023 (1.10 Ed.) , 2. The Linux Programming Interface A Linux and UNIX System Programming Handbook by Michael Kerrisk ,No Starch Press , 3. Computer Systems: A Programmer's Perspective by Randal E. Bryant & David R. O'Hallaron ,Pearson |

**DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:**

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

| S.no | Advanced Topics, Additional Reading, Research papers and any | CO | ALM | References/MOOCS |
|---|---|---|---|---|

**EVALUATION PLAN:**

| Evaluation Type | Evaluation Component | Weightage/Marks | | Assessment Dates | Duration (Minutes) | CO1 | CO2 | CO3 | CO4 | CO5 | CO6 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| End Semester Summative Evaluation Total= 40 % | Term End Project Defence & Demonstration | Weightage | 16 | | 90 | 3 | 2.5 | 2.5 | 2.5 | 2.5 | 3 |
| | | Max Marks | 90 | | | 15 | 15 | 15 | 15 | 15 | 15 |
| | End Semester Exam | Weightage | 24 | | 180 | 2 | 2 | 2 | 6 | 6 | 6 |
| | | Max Marks | 90 | | | 5 | 5 | 5 | 25 | 25 | 25 |
| In Semester Formative Evaluation Total= 36 % | Active Learning Engagement (In-Class) | Weightage | 6 | | 90 | 1 | 1 | 1 | 1 | 1 | 1 |
| | | Max Marks | 60 | | | 10 | 10 | 10 | 10 | 10 | 10 |
| | Practical Continuous Evaluation | Weightage | 9 | | 90 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |
| | | Max Marks | 60 | | | 10 | 10 | 10 | 10 | 10 | 10 |

| Evaluation Type | Evaluation Component | Weightage/Marks | | Assessment Dates | Duration (Minutes) | CO1 | CO2 | CO3 | CO4 | CO5 | CO6 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Project Continuous Evaluation | Weightage | 9 | | 90 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |
| | | Max Marks | 60 | | | 10 | 10 | 10 | 10 | 10 | 10 |
| | Hackathon | Weightage | 12 | | 240 | 2 | 2 | 2 | 2 | 2 | 2 |
| | | Max Marks | 100 | | | 16 | 16 | 17 | 17 | 17 | 17 |
| In Semester Summative Evaluation Total= 24 % | In-Term Practical Examination | Weightage | 6 | | 90 | 1 | 1 | 1 | 1 | 1 | 1 |
| | | Max Marks | 60 | | | 10 | 10 | 10 | 10 | 10 | 10 |
| | Mid-Term Examination (Descriptive) | Weightage | 12 | | 90 | 4 | 4 | 4 | | | |
| | | Max Marks | 45 | | | 15 | 15 | 15 | | | |
| | MOOCs Evaluation (Certificate + Test) | Weightage | 6 | | 90 | 1 | 1 | 1 | 1 | 1 | 1 |
| | | Max Marks | 60 | | | 10 | 10 | 10 | 10 | 10 | 10 |

**ATTENDANCE POLICY:**

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course
In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

**DETENTION POLICY :**

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

**PLAGIARISM POLICY :**

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

**COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:**

| Name of Faculty | Delivery Component of Faculty | Sections of Faculty | Chamber Consultation Day (s) | Chamber Consultation Timings for each day | Chamber Consultation Room No: | Signature of Course faculty: |
|---|---|---|---|---|---|---|
| Dyuti Banerjee | P | 13-B | - | - | - | - |
| Y Devi | L | 5-MA,9-MA | - | - | - | - |
| Y Devi | P | 9-A,5-A | - | - | - | - |
| Y Devi | S | 1-B,9-A,5-A | - | - | - | - |

| Name of Faculty | Delivery Component of Faculty | Sections of Faculty | Chamber Consultation Day (s) | Chamber Consultation Timings for each day | Chamber Consultation Room No: | Signature of Course faculty: |
|---|---|---|---|---|---|---|
| MUNIRAJU VADLAMUDI | L | 8-MA | - | - | - | - |
| MUNIRAJU VADLAMUDI | P | 8-A,1-B,5-B | - | - | - | - |
| MUNIRAJU VADLAMUDI | S | 8-A,3-B | - | - | - | - |
| Hema Kolla | L | 7-MA,3-MA | - | - | - | - |
| Hema Kolla | P | 7-A,3-A | - | - | - | - |
| Hema Kolla | S | 5-B,7-A,3-A | - | - | - | - |
| khaja sk | S | 12-B | - | - | - | - |
| DHULIPALLA KRISHNA | P | 6-B | - | - | - | - |
| Anitha Chekuri | S | 12-B | - | - | - | - |
| LakshmiPrasanna Yeluri | P | 12-B | - | - | - | - |
| Swathi D | P | 8-B | - | - | - | - |
| Swathi D | S | 3-B | - | - | - | - |
| Karumuri Murthy | P | 3-B | - | - | - | - |
| Gayatri Mantri | S | 10-B | - | - | - | - |
| Kotte Anusha | L | 6-MA,2-MA | - | - | - | - |
| Kotte Anusha | P | 6-A,2-A | - | - | - | - |
| Kotte Anusha | S | 6-A,8-B,2-A | - | - | - | - |
| borra sivaiah | P | 10-B | - | - | - | - |
| borra sivaiah | S | 12-B | - | - | - | - |
| DHARAMSOTU BHEEKYA | L | 1-MA | - | - | - | - |
| DHARAMSOTU BHEEKYA | P | 7-B,4-B,1-A | - | - | - | - |
| DHARAMSOTU BHEEKYA | S | 7-B,1-A,4-B | - | - | - | - |
| paramewar maddela | P | 11-B | - | - | - | - |
| paramewar maddela | S | 13-B | - | - | - | - |
| pavani Chilumula | P | 9-B,2-B | - | - | - | - |
| pavani Chilumula | S | 2-B,6-B,9-B | - | - | - | - |
| Dasari . | L | 13-MA,11-MA | - | - | - | - |
| Dasari . | P | 13-A,11-A | - | - | - | - |
| Dasari . | S | 13-A,11-A | - | - | - | - |

| Name of Faculty | Delivery Component of Faculty | Sections of Faculty | Chamber Consultation Day (s) | Chamber Consultation Timings for each day | Chamber Consultation Room No: | Signature of Course faculty: |
|---|---|---|---|---|---|---|
| Raghupathi Manthena | L | 10-MA,12-MA | - | - | - | - |
| Raghupathi Manthena | P | 12-A,10-A | - | - | - | - |
| Raghupathi Manthena | S | 10-A,12-A | - | - | - | - |
| Soumya Enukonda | L | 4-MA | - | - | - | - |
| Soumya Enukonda | P | 4-A | - | - | - | - |
| Soumya Enukonda | S | 11-B,4-A | - | - | - | - |
| MEKALA SANJEEV KUMAR | S | 11-B,13-B | - | - | - | - |

**GENERAL INSTRUCTIONS**

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

**NOTICES**

All course-related notices and communications will be shared through the official institutional email.

**Signature of COURSE COORDINATOR**

(Kotte Anusha)(9985)(CSE-1)

**Signature of Department Prof. Incharge Academics & Vetting Team Member**

Department Of CSE-1

**HEAD OF DEPARTMENT:**

**Approval from: DEAN-ACADEMICS**
(Sign with Office Seal)
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